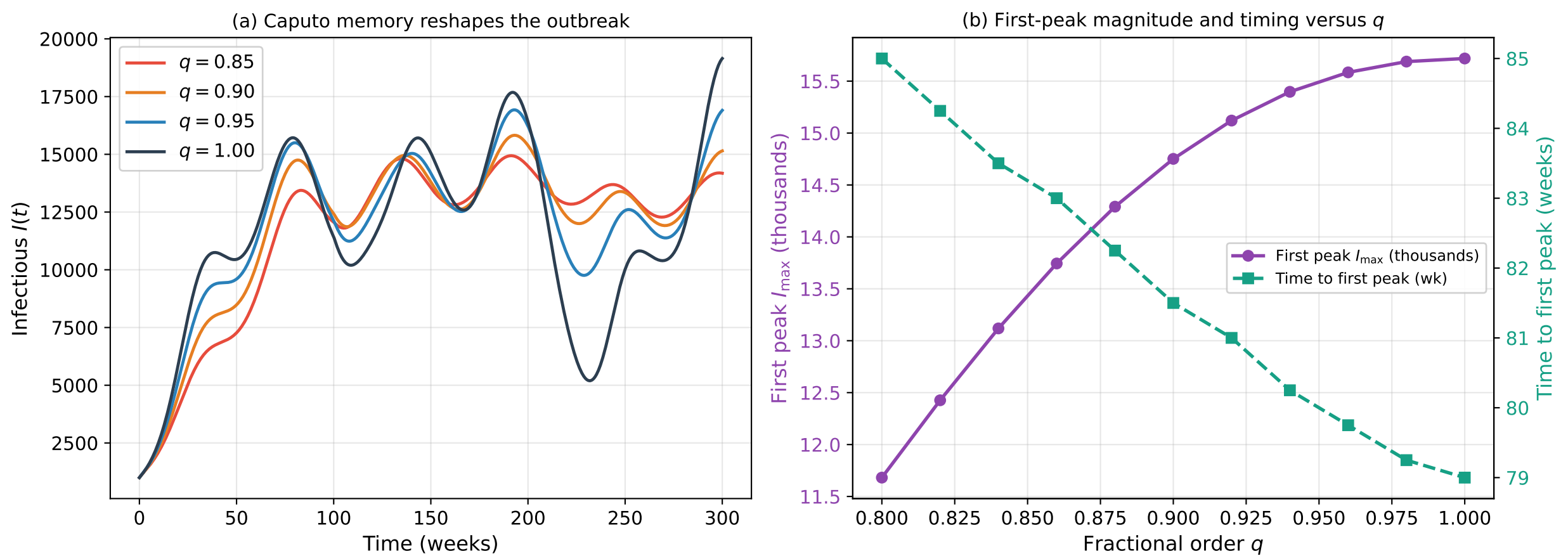
(a) Caputo memory reshapes the outbreak
Infectious $I(t)$
Time (weeks)
$q = 0.85$
$q = 0.90$
$q = 0.95$
$q = 1.00$
20000
17500
15000
12500
10000
7500
5000
2500
0
50
100
150
200
250
300
(b) First-peak magnitude and timing versus q
First peak I_{max} (thousands)
Time to first peak (weeks)
Fractional order q
First peak I_{max} (thousands)
Time to first peak (wk)
15.5
15.0
14.5
14.0
13.5
13.0
12.5
12.0
11.5
85
84
83
82
81
80
79
0.800
0.825
0.850
0.875
0.900
0.925
0.950
0.975
1.000